

Masa

Data voice

How to use this: not a verbatim script — nobody is expected to read from it word-for-word, and realistically no one will stick to a script anyway. Study it well enough to explain each point in your own words, improvise if you're cut off, and answer a follow-up without freezing. If a question lands outside your section, give the short version and hand off by name — every presenter's script and the underlying `FIELD-NOTES.md` are in the same GitHub repo.

日本語 (JAPANESE)

あなたはデータ担当です。あなたの担当スライドにはすべてグラフ、または費用の積算表があります。数字を自分のものとして扱い、はっきりとゆっくり話し、各スライドの核心となる数字を急いで通り過ぎないようにしてください。予算の内訳スライド(i1~i3)があなたの担当なのは、それが説得ではなく計算だからです。すべての項目は出典付きの市場価格から積算されています。内訳スライドは本編の直後(i1~i3)にあり、ケイスケのスライド14の金額タイルをタップすると直接ジャンプします。求められたらその場で説明できるようにしておいてください。

Field-visit recap (read this first if you missed a visit)

Everything in the deck traces back to these visits — this is the shared ground truth every presenter should know, whether or not you were there.

Tatay's cacao farm, Batbat (Tuesday, Aug 26). A 2.5-hectare farm run by Tatay alone — his children work in the cities — 100% organic, and he has health problems that limit how much heavy labor he can do himself. His farm used to produce about 500 kg of cacao a year (sold at ₱75–100/kg); this season it's close to **zero**, because of pests and disease. His own estimate of the farm's potential if pests were solved: **2,000 kg/year** — a 4× recovery, the biggest single lever we found anywhere. Cacao flowers year-round here, so the pest pressure never gets an off-season. On water, his framing was specific: the constraint is water *pressure*, not volume — he can't even estimate how many sprinklers his source could feed — and he's open to drip irrigation and rainwater collection. Important attribution point: Tatay described the problems and constraints; the pilot plan, the tools, and every proposed solution came from our own investigations, not from him.

What the farmers and Muravah told us directly. Pest control is their #1 problem — ahead of typhoon, which everyone called "unpredictable" and would rather we not chase. Post-harvest losses run 80–100% from pests/disease across the network, and 60% of pods die from careless handling alone (worse under Mayon ashfall). Farmer income is ₱2,000–10,000/month — below Bicol's ₱13,624/month poverty line for a family of five — the average farmer is about 50, and there's no succession pipeline; Muravah copes by

recruiting BU Guinubatan students. Two hard rules for any solution: **organic only** (no synthetic fertilizers or pesticides, ever) and **no AI/robotics** ("not there yet," "AI is biased according to the natives") — low-tech and human-in-the-loop is a values position for them, not a budget issue. More labor doesn't raise profit, because pest pressure caps yield no matter how much work goes in.

Muravah Foundation / Mayon Gold. The cacao NGO anchoring all of this: cacao profits fund typhoon-proof homes and scholarships, heirloom native cacao interplanted with Pili and coconut, and a production-cycle data study already running across 500–1,000 farmers — which is where our pilot would live.

The processing facility. Their own #1 problem is power outages, but the line that matters for us: they're "sometimes out of supply for cacao." Farm-level pest losses upstream directly destabilize the factory downstream — fixing the farms fixes two problems at once.

The other farms (Aug 27–28, now Extras slides). Tatay Aldavis's rice/dairy farm school (a different "Tatay" — it's a Bikol honorific for an elder, not a name) controls pests with herbal sprays and ducks that eat snails — proof organic pest control works in Bicol; plus a bamboo farm, stingless-bee honey producers, the Zambales agrivoltaics site (the ₱10M funding benchmark), and a youth agritourism program.

Your slides

You're the data voice. Every one of your slides has a chart or a costed ledger — own the numbers, say them clearly and slowly, and don't rush past the punchline number on each slide. The three budget breakdowns (i1–i3) are yours because they're arithmetic, not persuasion: every line is priced from a cited market source. They sit right after the thank-you and are reached by tapping the budget tiles on Keisuke's slide 14 — be ready to walk one on demand.

Slide 6 — Path to finished cacao: direct vs. stored.

Muravah's stated production timeline: flowering to pod-ready takes **6 months**, then ferment **6 days**, solar-dry **1 day to a week**, roast **27 minutes at 140°C**. The chart shows two paths to a sellable product: processing right after drying gets you to **day 191**; storing dried beans 3–6 months before processing (a real risk — delay invites humidity damage) roughly doubles that to **day 326**. A third line shows a **benchmark from the cocoa literature: a well-run farm is sellable by ~day 178** [1][2] — our direct path is within two weeks of it, which supports the claim that the process itself is sound and the losses are the problem. Also worth mentioning: the facility is undersized for its own batch volumes (pod-breaking up to 1,000 pods/hour at peak), 60% of pods are lost to careless handling (worse with Mayon ashfall), and — an important framing point — **more labor doesn't raise profit here**, because pest/disease pressure caps yield regardless of how much work goes in. That last point is why treating pests/disease is the highest-leverage fix — the through-line of the whole deck.

ムラウ財団が示した生産タイムラインは次の通りです。開花からポッド(実)の収穫までが6か月、発酵に6日、天日乾燥に1日から1週間、そして140℃で27分間のロースト(焙煎)です。グラフは製品化までの2つの経路を示しています。乾燥後すぐに加工すれば191日目で完成しますが、加工前に乾燥豆を3〜6か月保管すると(保管中に湿気で傷むリスクがあります)、326日目とほぼ倍になります。3本目の線はカカオ文献のベンチマークで、管理の良い農園は約178日目で販売可能になります(出典1・2)。私たちの「直接加工」経路はこのベンチマークと2週間しか違いません。つまりプロセス自体は健全で、問題は損失の方だという主張の裏付けになります。また、施設自体のバッチ量に対して規模が小さいこと(ピーク時には1時間に1,000個ものポッドを割る作業が必要)、丁寧に扱わなければポッドの60%が損失すること(マヨン山の降灰でさらに悪化)も触れる価値があります。そして重要な論点として、ここでは労働量を増やしても利益は増えません。なぜなら病害虫の圧力が労働量にかかわらず収量に上限をかけてしまうからです。この最後の点こそ、病害虫対策への投資が最も効果の高い打ち手である理由であり、このプレゼン全体を貫くテーマです。

Slide 13 — Why this works (feasibility).

This slide was flagged as especially important — don't rush it.

1. Everything proposed is organic-only, low-cost, human-in-the-loop, no synthetic inputs or AI/robotics — that's not a technology choice, it's literally what Muravah and Tatay told us they need.
2. It's the single largest lever in the dataset: resolving pest/disease pressure is a ~4× yield recovery for Tatay — 0 kg to a potential 2,000 kg/year.
3. It fixes two problems at once — stabilizing farm-level supply also stabilizes the processing facility, which told us it's "sometimes out of supply for cacao" because of these same upstream losses.
4. The ideas are properly researched — every measure is cross-checked against published research and priced from market data, not invented from scratch. The clickable citations and the References page at the end of the deck are the proof; point at them if challenged.

このスライドは特に重要だと指摘されています。急がずに丁寧に説明してください。 1. 提案する内容はすべてオーガニックのみ、低コストで、人が判断に関わる仕組みであり、合成資材やAI・ロボティクスは使用しません。これは技術的な好みではなく、ムラワ財団とタタイさん自身が必要だと語った内容そのものです。 2. データセット全体で最大の改善余地です。病虫害の圧力を解消できれば、タタイさんの収量は約4倍に回復します、0kgから潜在的な2,000kg/年へ。 3. 2つの問題を同時に解決します。農場レベルの供給が安定すれば、「カカオの供給が時々不足する」と語っていた加工施設側の問題も、同じ上流の損失が原因であるため、あわせて安定します。 4. 提案内容はすべてきちんと調査・裏付けされています。各施策は公表された研究と照合し、費用は実際の市場価格から積算しています。ゼロからの発明ではありません。スライド上のクリックできる出典番号と、最後の参考文献ページがその証拠です。疑問を持たれたら、その場で示してください。

Slide i1 — Where the ₱100K starter figure comes from (via the slide-14 tiles).

One farm, one year, at 2025–26 Philippine market prices, now grouped with subtotals so the shape is readable at a glance — say the three subtotals, not every peso:

- **Water system, subtotal ₱52,000:** drip starter zone 0.25 ha ₱30,000 [13]; two 1,000 L tanks plus stand ₱14,000 [12]; Raspberry Pi controller kit with sensors, relay and valves ₱8,000 [14].
- **Pest & disease control, subtotal ₱17,000:** biocontrol starter colonies — carnivorous snails that hunt the pest snails, plus the weaver ants and lynx spiders PCAARRD mass-rears — ₱5,000 [15]; a year of bio-spray inputs ₱12,000 [1].
- **Program & people, subtotal ₱31,000:** printed calendars ₱4,000; twice-monthly monitoring visits ₱15,000 [11]; training day + ~15% contingency ₱12,000.
- **Total ≈ ₱100,000.** Every line has a cheaper local substitute; the drip zone and the controller kit are the only equipment purchases.
- **The expected-return panel sits on the right of the slide — read it with the total:** with pests resolved the farm yields up to 2,000 kg/yr, and at the ₱75–100/kg farm-gate price Tatay quoted that's **₱150K–200K of cacao a year** [1] — payback within the first full harvest, and even a half recovery covers the ₱100K in about a year.

日本語 (JAPANESE)

1つの農場、1年間、2025～26年のフィリピン市場価格での積算です。表は小計付きのグループ形式になったので、1ペソ単位ではなく3つの小計を読み上げてください。水システム小計5.2万ペソ:点滴スターターゾーン3万(出典13)、タンク2基と架台1.4万(出典12)、Piコントローラーキット8千(出典14)。病虫害対策小計1.7万ペソ:生物防除の導入コロニー(害虫のカタツムリを捕食する肉食性カタツムリ、ツムギアリ、ハシリグモ)5千(出典15)、1年分のバイオスプレー資材1.2万(出典1)。プログラム・人件費小計3.1万ペソ:印刷物4千、月2回の巡回1.5万(出典11)、研修と予備費1.2万。合計約10万ペソです。スライド右側の「期待リターン」パネルを合計と一緒に伝えてください:病虫害が解決すれば収量は最大2,000kg/年、農場渡し価格75～100ペソ/kgで年間15万～20万ペソのカカオ(出典1)。回収は最初の本格収穫内、回復が半分でも約1年で元が取れます。

Slide i2 — Where the ~P1M for ten farms comes from.

Scaling the starter pilot inside Muravah's existing study — the key sentence is that **shared costs don't multiply by ten**. Three subtotals: **water P520,000** (ten drip zones P300K [13], twenty tanks P140K [12], ten Pi kits P80K [14]); **pest & disease P160,000** (biocontrol colonies for ten farms P40K [15], bio-spray P120K [1]); **program & people P350,000** (printing and trainings P45K, coordinator and student program P180K [11], data materials P40K, contingency P85K) — **total ≈ P1,030,000, headline ~P1M**. The expected-return panel on the right: if the ten farms average Tatay's recovery that's up to 20,000 kg/yr, worth **P1.5–2M a year** at farm-gate prices [1] — payback inside the first full harvest year, before counting any organic price premium.

日本語 (JAPANESE)

ムラワ財団の既存調査の中でスターターパイロットを10農場に拡大した場合の積算です。鍵となる一文は「共有できる費用は10倍にはならない」。3つの小計で伝えてください。水システム小計52万ペソ:点滴ゾーン10か所30万(出典13)、タンク20基14万(出典12)、Piキット10台8万(出典14)。病虫害対策小計16万ペソ:10農場分の生物防除コロニー(肉食性カタツムリ、ツムギアリ、ハシリグモ)4万(出典15)、バイオスプレー12万(出典1)。プログラム・人件費小計35万ペソ:印刷と研修4.5万、コーディネーターと学生プログラム18万(出典11)、データ資材4万、予備費8.5万。合計約103万ペソ、見出しは約100万ペソです。右側の期待リターンパネル:10農場がタタイさんの回復水準に達すれば最大20,000kg/年、年間150万～200万ペソのカカオ(出典1)。回収は最初の本格収穫年内、オーガニックのプレミアム価格はその上乗せです。

Slide i3 — The P10M benchmark isn't ours to estimate.

Be precise about what this number is: P10,000,000 is what the Zambales agrivoltaics project **actually raised**, as cited to us on the site visit [1] — it is not our costing. Our one-farm ask (P100K) is **1%** of that proven benchmark, and the ten-farm ask (~P1M) about **10%**. That's the point of the slide: the region has already moved money at this scale, and we're asking for a fraction of it. Returns framing: unlike

infrastructure spending, our ask is harvest-backed — ten recovered farms sell ₱1.5–2M of cacao a year, roughly 10% of this benchmark sum returned annually from produce alone [1]. Full scale could later add what the agrivoltaics precedent already pairs with farming — solar-powered irrigation and cold storage [10].

日本語 (JAPANESE)

この数字が何であることを正確に伝えてください。1,000万ペソは私たちの見積もりではなく、ザンバレスのアグリボルタイクス(営農型太陽光)事業が実際に調達した金額として、現地訪問で私たちに示された数字です(出典1)。私たちの1農場の要望額(10万ペソ)はこの実績あるベンチマークのちょうど1%、10農場(約100万ペソ)でも約10%に過ぎません。これがこのスライドの主張です:この地域はすでにこの規模の資金を動かした実績があり、私たちの提案はその一部で済みます。リターンの位置づけ:インフラ投資と違い、私たちの提案は収穫に裏付けられています。回復した10農場は年間150万~200万ペソのカカオを販売でき、このベンチマーク額の約10%を毎年、農産物だけで返します(出典1)。将来的な本格展開では、アグリボルタイクスの前例がすでに農業と組み合わせているもの、太陽光発電による灌漑とコールドストレージを追加できます(出典10)。

Slide e3 — What the Zambales solar precedent tells us (3 charts, Extras).

1. *Panel efficiency* degrades gradually over a 30-year lifespan — 100% new down to about **82.5% at year 30** (some owners replace panels early, at 10–15 years).
2. *Panel waste, by weight*: 75% glass, 15% aluminum, 10% other electronics/polymers — relevant to anyone's lifecycle-cost math for a solar-powered irrigation proposal.
3. *Panel orientation*: an optimally tilted panel is the 100% baseline, flat gets about 85%, and a **vertically mounted panel only about 65%**. Say clearly that this last chart is our own general engineering estimate, not a cited interview figure — it sets up Keisuke's callout on e4.

日本語 (JAPANESE)

1. パネルの発電効率は30年の耐用年数の間に緩やかに低下します。新品時の100%から30年目には約82.5%まで低下します(所有者によっては10~15年で早期に交換する場合があります)。 2. パネル廃棄物の重量内訳:ガラスが75%、アルミニウムが15%、その他の電子部品・ポリマーが10%です。これは太陽光発電を利用した灌漑提案のライフサイクルコストを考える上で重要な情報です。 3. パネルの設置角度:最適な角度で設置した場合を100%の基準とすると、水平設置では約85%、そして垂直(壁面)設置ではわずか約65%程度になります。この最後のグラフは、インタビューで得られた数値ではなく、私たちチーム独自の一般的な工学的推定値であることを明確に伝えてください。ケースのe4の指摘につながります。